SixRing™

A Chemical Evolution™ Company

The SixRing™ Solution

Scalable + Economic + Sustainable = SixRing

Patented Delignification Chemical Process

Biomass

#1 Feedstock

SixRing Feedstock is waste, non-food biomass:

- Low cost
- Abundant
- Scalable & Flexible
- Processed 50+ feedstocks, including blends

#2 SixRing Process

Ambient Temperature and Pressure

- Lower CAPEX
- Lower OPEX
- Lower Carbon Intensity
- Higher Yields
- Unique & Valuable Products

#3 Product Streams

High Value

High Volume

#4 *Saleable Products*

Calgary + Fulton Facility Focus

Agriculture + Biostimulants

- High-Value Streams Retained
- 100% Owned Production Assets
- High Dilution Factor = High Margin
- Sustainable Replacements
- ***SixRing Leading Solution***

Energy – Drop-in Fuels, RNG, etc.

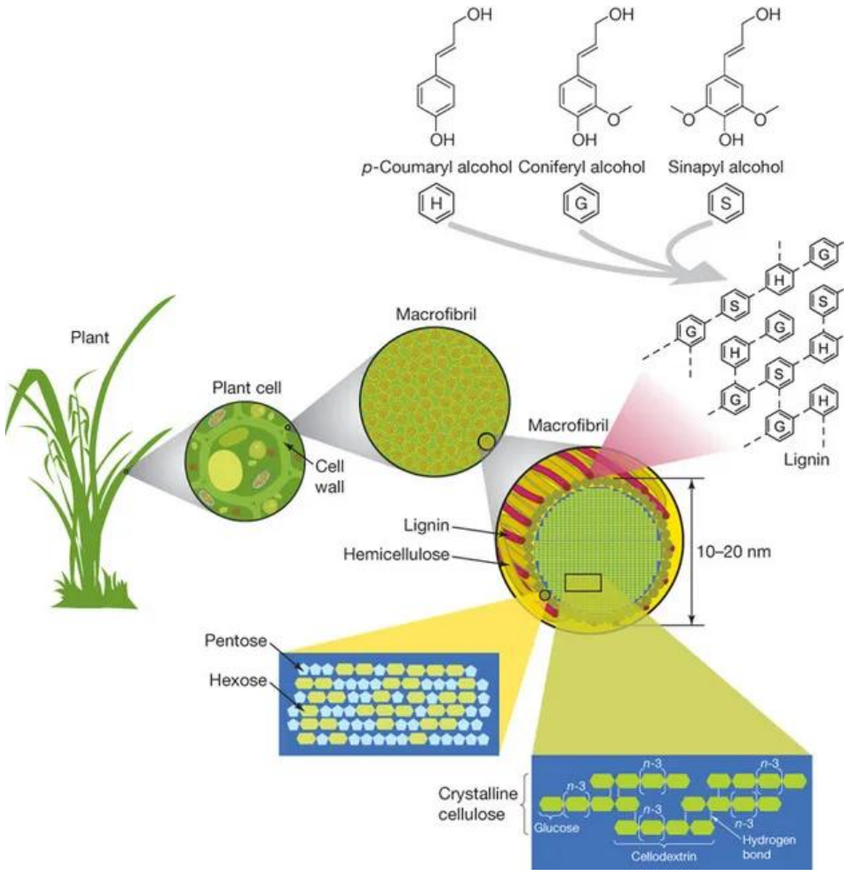
- High-Volume Fuel Pathways
- Capital-Light Licensing Model
- Well-Capitalized Partners
- Scalable Deployment

#5 *Project & Offtake Partners*

Executed agreements with counterparties

Non-Food Biomass Feedstock

What is Lignocellulosic Biomass?



Three Elements of Biomass

1 Lignin

Protective, very durable, and biocidal outer layer

2 Cellulose

Inner layer that is protected by the lignin

3 Hemicellulose

Branched, linking material

Unlocking Value in Biomass

The elements of the biomass (lignin, cellulose, and hemicellulose) are **each known to be highly valuable**

The problem has always been **difficulty in separating the elements without destroying value**

Incumbent technologies use high pressure and temperature to destroy the lignin, resulting in lower yields, and strained economics

Incumbent Processing Technologies

Proprietary & Patented Biomass Process

SixRing's **First-of-a-Kind, Feedstock Agnostic, Chemical Solution** Overcomes Challenges Experienced by Existing Biomass Conversion Technologies

Incumbent Technologies

High Temperature and Pressure (1000C)	Low Temperature and Pressure (300C)
Gasification Pyrolysis Liquefaction	Hydrolysis Fischer-Tropsch

Current sustainable biomass projects use legacy technologies, which are over 100 years old.

Next Generation Technological Advancement

The SixRing Process

AMBIENT Temperature and Pressure (35C)

The **Ambient Process** allows SixRing to delicately separate each valuable element of the biomass (Lignin, Cellulose, and Hemicellulose), **resulting in three distinctly valuable product streams**, with better economics and better carbon intensity.

BioStimulants in Agriculture

Natural solutions for enhancing plant vigor and crop resilience

What are BioStimulants?

Natural substances or microorganisms that improve plant health, nutrient efficiency, and stress tolerance independent of direct nutrient supply.

Key benefits

- **Nutrient uptake:** Enhance efficiency and use of fertilizers.
- **Stress tolerance:** Resilience to drought, salinity, and temperature fluctuations.
- **Root growth:** Stimulation of early development and soil interactions
- **Crop Quality:** Improved colour, size, and stability.

Mechanisms of Action

- **Seed stimulation:** Enhance germination rates under stress by improving seed resilience.
- **Nutrient availability:** Chelation of micronutrients making them more bioavailable and improving photosynthesis.
- **Microbial stimulation:** Provide a carbon source for microbes leading to increased nutrient cycling and soil health.
- **Root growth:** Interact with plant signaling and trigger hormone-like effects leading to root elongation and branching.
- **Stress mitigation:** Modulate antistress responses by boosting the production of antioxidant molecules.

TeraCoat™ and TeraSix™ BioStimulants

Sustainable biomass-derived stimulation for natural plant development and growth

Benefits of TeraSix™ and TeraCoat™:

- **Natural origin:** Derived from the delignification of sustainable biomass at ambient temperatures and pressures.
- **100 % water soluble**
- **Highly efficient:** Small amounts per acre or applied to seeds will result in great results.
- **Optimized chemistry:** Both products are rich in carboxylic acids and phenol groups which are known to support seedling establishment and plant development.
- **Free of EDTA**
- **Thoroughly tested:** Our products are tested in SixRng's world class laboratories.
- **Compatibility with other products:** Due to its simple yet efficient formulation, TeraCoat™ and TeraSix™ are formulated to be compatible with other agricultural products.

How are TeraSix™ and TeraCoat™ different than Humic Acids and Lignosulfonates?

Although they share similar functional groups, carboxylic acids and phenolic hydroxyl groups important for plant growth and development, the structure of these groups are vastly different:

SixRing's TeraSix™ and TeraCoat™ have **smaller molecular weights** (500-2,000 Da - Daltons) and narrower molecular weight distributions than humic acids (5,000 to > 1M Da) allowing for a better interaction with the plant and a **higher functional group density** within the molecule.

TeraCoat Seed Testing Protocol

Black lentils

TeraCoat™ on black lentils

Protocol

- Treatments were placed in 72-insert trays.
- 24 replicates per treatment were used and placed in two rows in the 72-insert tray.
- One seed deployed / planted for each cell in order to calculate germination rate.
- Watering was conducted by an automatic mister for 1 minute each day at 6 am.
- Lights were set to 12 hours of daylight (6 am to 6 pm).
- Plants were harvested 2 weeks after seeding.

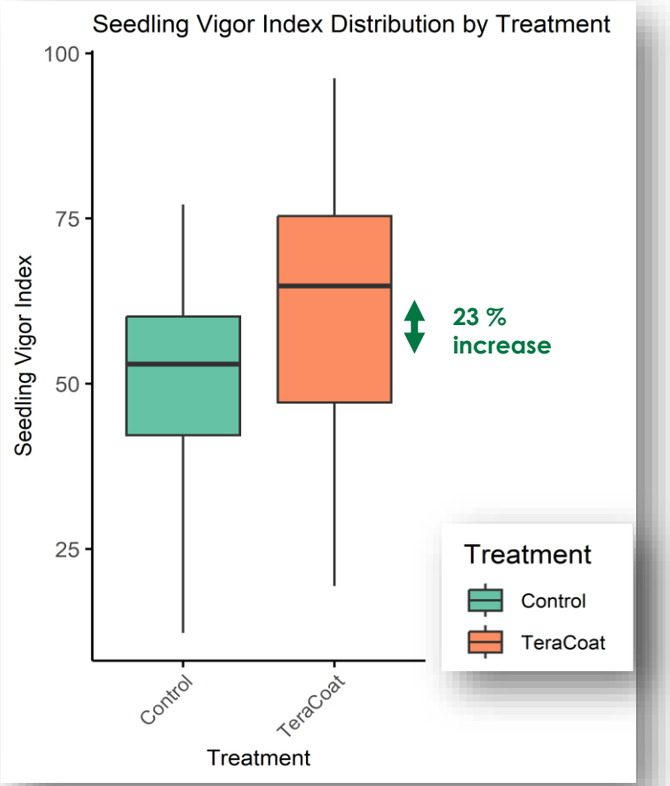
TeraCoat™

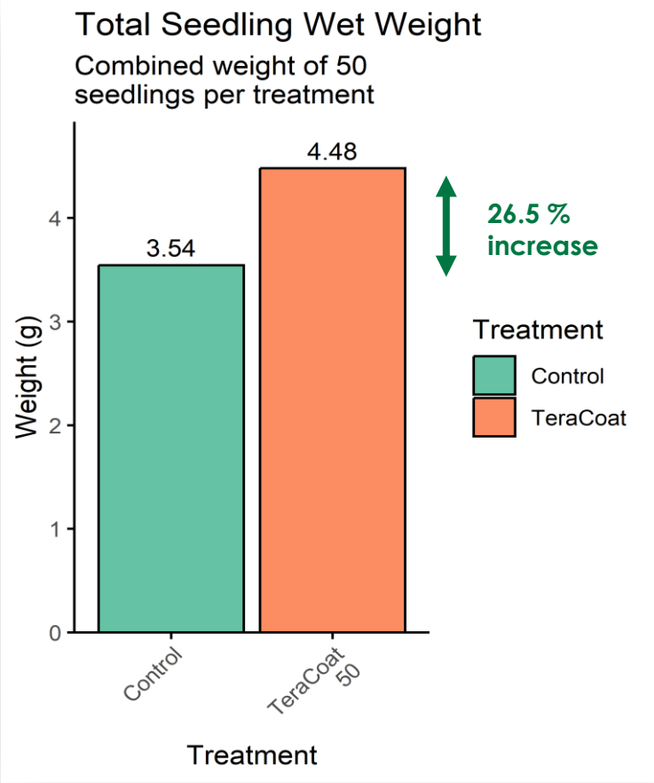
- At the time of harvest, plants were removed from trays and the soil was carefully washed for optimal root exposure for photos of the root and shoot development.
- Roots and shoots were measured to evaluate growth and development of seedling.

TeraCoat Seed Treatment

Black Lentils

Scientifically Significant Test Results (Black Lentil)





Showed a dramatic increased wet weight over the controls.

Showed increased germination statistics (days to germination and germination index).

Had a major increase in seedling vigor index.

TeraCoat Seed Treatment - Black Lentils

Sustainable biomass-derived stimulation for plant development and growth

Black Lentil Seeds treated with TeraCoat:

- Showed a visually significant increase in root structure compared to controls and major market leader.

Control

TeraCoat

Leading Competitor

TeraCoat Seed Testing Protocol

Green lentils & Maple peas

TeraCoat™ treatment on green lentils and maple peas

Protocol

- Treatments were randomly distributed on 50 insert plug trays with all treatments appearing on every tray.
- A total of 3 trays with 30 replicates per treatment (meaning 10 replicates per tray per treatment) were used.
- One seed deployed / planted for each cell in order to calculate germination rate.
- Watering was conducted by an automatic mister for 1 minute each day at 6 am.
- Lights were set to 12 hours of daylight (6 am to 6 pm).
- Plants were harvested 3 weeks after seeding.

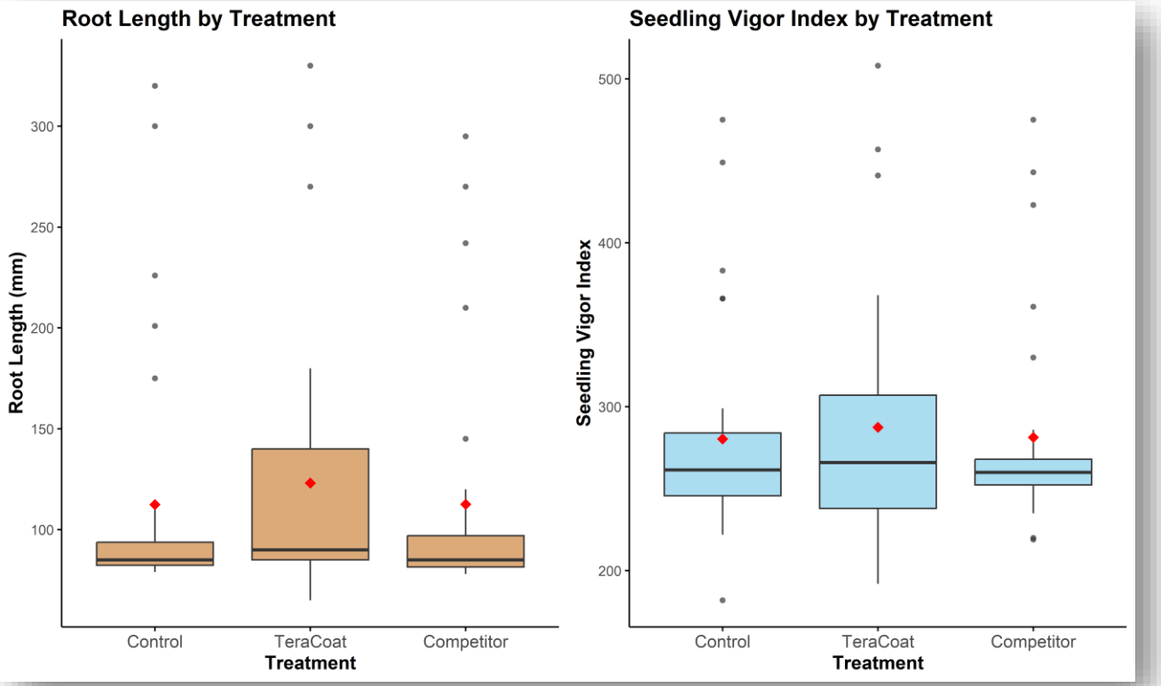
TeraCoat™

- At the time of harvest, plants were removed from trays and the soil was carefully washed for optimal root exposure for photos of the root and shoot development.
- Roots were then separated from the shoot for further weight measurements.

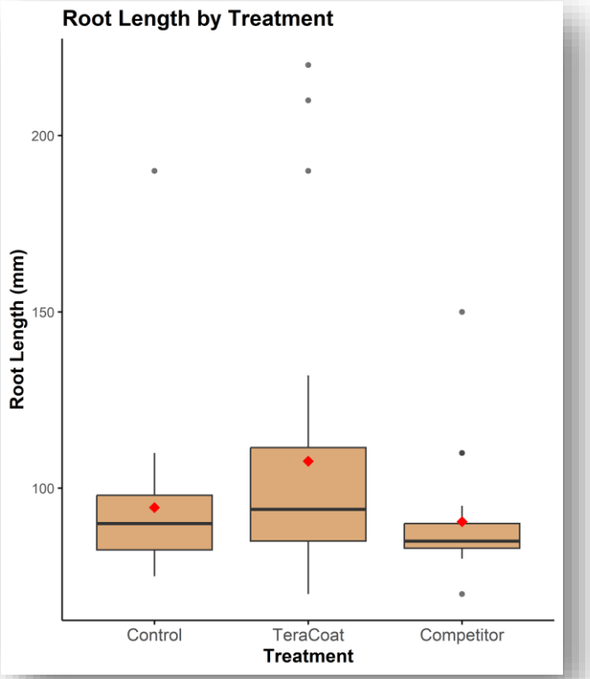
TeraCoat Seed Treatment

Green lentils & Maple peas

Scientifically Significant Test Results (Green Lentils & Maple Peas)



Maple peas



Green lentils

Considerable increase in root length

Had an increase in seedling vigor index

Box-and-whisker plot showing the middle of the data as a bold line, the overall spread within the box, and any unusual values as dots. The red diamond signifies the average.

Industry Leading Performance at Competitive Pricing (Maple Peas)

TeraCoat™

Leading Competitor

Control

Industry Leading Performance at Competitive Pricing (Green Lentils)

TeraCoat™ | **Leading Competitor** | **Control**

Leading Fertilizer Compatibility Experimentation (Maple Peas)

TeraCoat™

Leading competitor

Control

- Treatments were randomly distributed on insert plug trays with all treatments appearing on every tray (6 trays, 30 replicates).
- Watering was conducted by an automatic mister for 1 minute each day at 6 am. Lights were set to 12 hours of daylight (6 am to 6 pm).
- Plants were harvested 2 weeks after seeding. Samples were treated with a mixture of TeraCoat™ with market fertilizer.

How Does SixRing Compare?

SixRing: A New Category of High-Performance Biostimulants

Natural, Low-Cost Biostimulants with Hormone-Like Signaling and Scalable Production

	Lignin Products	Plant Extracts	Plant Hormones	SixRing Products
Stimulant Types:	Humic Acids Lignosulfonates	Seaweed Extract Amino Acid Hydrolysates	Auxins Cytokinin Gibberellin	**TeraCoat (Seed Coating)** **TeraSix (Foliar)**
Molecular Weight & Plant Signaling:	**High molecular weight:** No plant signaling activity	**Low molecular weight:** Could act as plant signaling molecules	**Low molecular weight:** Direct plant hormone signaling	**Low molecular weight:** Acts as plant signaling molecule to stimulate plant growth
Concentration of Active Ingredients:	**Low concentration:** Many inert molecules due to manufacturing process	**Low concentration:** Inert molecules & limited active ingredient concentration	**High concentration:** Produced as purified hormone	**High concentration:** Ultra pure active ingredients (phenolics, aromatics, carboxylic acids)
Natural vs Synthetic:	**Natural / Mixed:** Humic + Lignosulfonates	**Natural:** Plant-derived extracts	**Synthetic:** Chemically produced hormones	**Natural:** Derived from lignin (non-food plant)
Price & Scalability:	**Low price & low performance:** Primarily soil conditioning	**Medium price & low performance:** Limited active ingredient concentration	**Very high price (not scalable):** Synthetic, regulated, and difficult to scale	**Low price & high performance:** Highly scalable natural biostimulant (EDTA free)